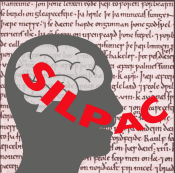


Learnability in Diachrony

Carola Trips (University of Mannheim)

Ringvorlesung: Unwrapping the mind: knowledge cognition and AI
HWS 2025

October 22, 2025

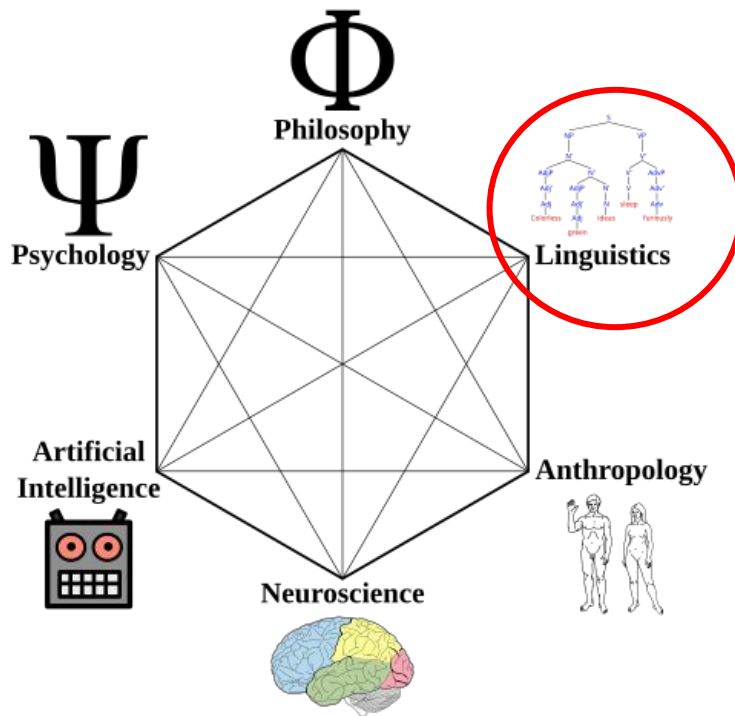


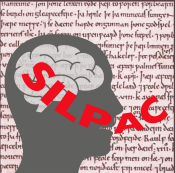
Linguistics: language acquisition and language change

Cognitive science: the study of the mind and its processes.

Linguistics:

- language as mental faculty
- language acquisition $\longleftrightarrow$ language change
- language learning

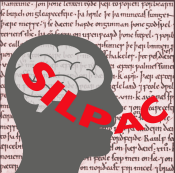




Linguistics: language acquisition and language change

Some questions to you to start up with:

1. Did humans 1,000 years ago have the same cognitive capacity to acquire language as humans today?
2. Do children drive language change?
3. Can intense contact with another language change the structure (grammar) of language?



Chair Anglistik IV: Linguistics/Diachrony



Dr. Juliane Elter



Lena Kaltenbach, MA
(SILPAC)



PD Dr. Michael
Percillier



Dr. Tara Struik
(SILPAC)



Alissa Selivenko, MA
(SILPAC)



PD Dr. Katrin Menzel

- Language change
- Language contact
- Syntax, morphology
- Linguistic theory
- Corpus linguistics

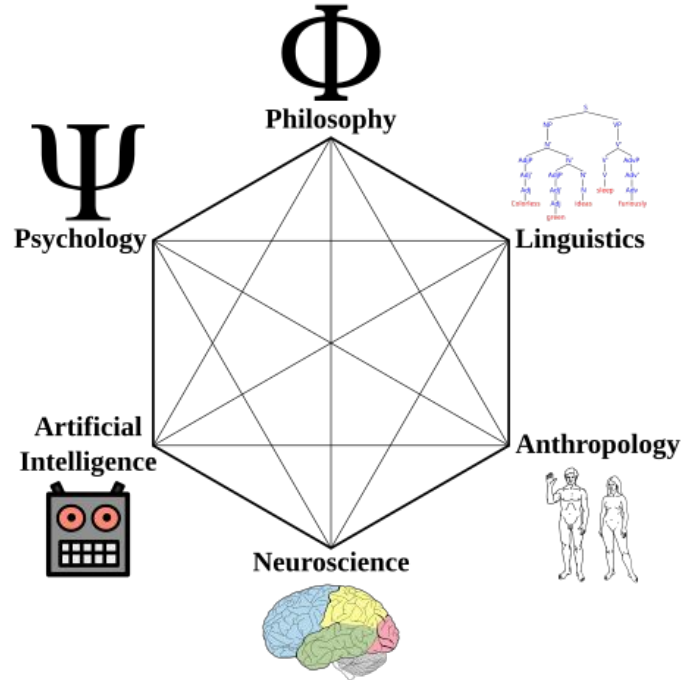


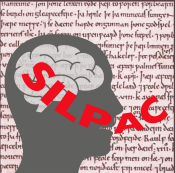
Outline

1. Links between psychology, psycholinguistics, and historical linguistics
2. Historical linguistics
 - a. Aims and objectives
 - b. Methods
 - c. Case study of labiality and contact-induced change
3. Psycho-historical linguistics: A new way to do historical linguistics
4. Future perspectives

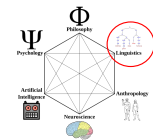
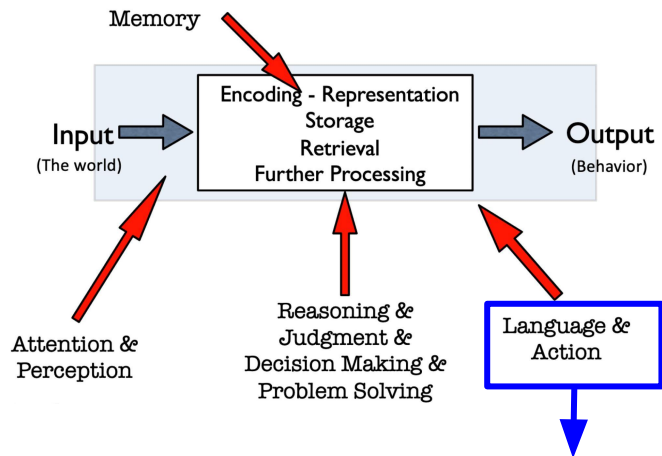


Psychology, psycholinguistics, historical linguistics as cognitive sciences

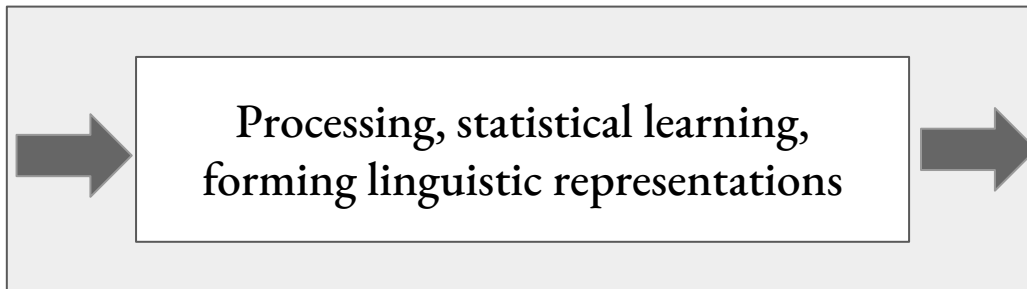




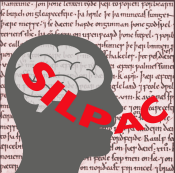
Psychology, psycholinguistics, historical linguistics as cognitive sciences



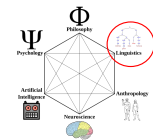
**Linguistic
input
(learner is
exposed to)**



**Linguistic
output
(grammar)**



Psychology, psycholinguistics, historical linguistics as cognitive sciences



	Experimental	Corpus-based	Children	Adults	Diachronic	Quantitative
Psychology, psycholinguistics	✓	—	✓	✓	—	✓
Historical linguistics	—	✓	—	✓	✓	✓



1. Historical linguistics





1. Historical linguistics

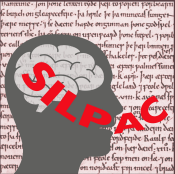
Historical linguistics

A branch of linguistics which studies the development of language and languages over times; also known as diachronic linguistics. Its primary goals are:

1. **Tracing language relationships** (e.g. the Indo-European language family)
2. **Reconstructing Proto-languages** (using the comparative method it reconstructs older stages of languages that are not attested)
3. **Explaining language change** (mechanisms of change, e.g. sound shift, changes in the syntax, lexicon, etc.)



(Campbell, Lyle. 2004. Historical linguistics: an introduction. Edinburgh: Edinburgh University Press)



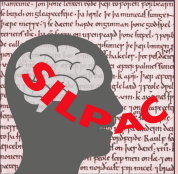
1. Historical linguistics: language acquisition and language change

Middle English (1150-1500)

- (1) With that **Com_v Merlion_s** on a grete blacke horse.
with that came Merlin on a great black horse
"At that moment **Merlin_scame_v** on a great black horse."
(PPCME2, M4, CMMALORY,28.893)
- (2) We fynd in profycy þat þe romans schvld **vs_o destroy_v**.
we find in prophecy that the Romans should us destroy
"We find in prophecy that the Romans should **destroy_v us_o**."
(PPCME2, M4, CMSIEGE, 86.512)



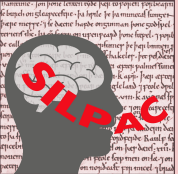
Where do these changes "come from"? Who actuates them? Look at acquisition!



1. Historical linguistics: language acquisition and language change

=> children successfully attain a "correct target grammar" from a class of available target grammars with little input and a lot of variation

Children acquire a strikingly uniform grammar within a speech community: "The end result is a high degree of uniformity in both the categorical and variable aspects of language production, where individual variation is reduced below the level of linguistic significance" (Labov 2012, 265; see also Labov 1972) (from Yang 2023: 1)



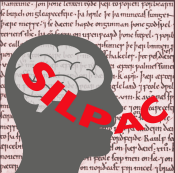
1. Historical linguistics: language acquisition and language change

"This [the logical problem of language acquisition] leads to an apparent paradox, i.e., the **logical problem of language change**. After all, if all children **successfully** attain the grammars of their parents and they continue to do this generation after generation, then the linguistic composition of every generation would look exactly like the linguistic composition of the previous generation and languages would not change with time. Yet, they do. So for languages to change with time, it must be the case that at least some children do not attain the grammar of their parents. It is these children, the mis-learners if you will, who presumably drive language change."

(Niyogi and Berwick, 1998, 192f)



This is why historical linguists need to have an understanding of acquisition and learnability.



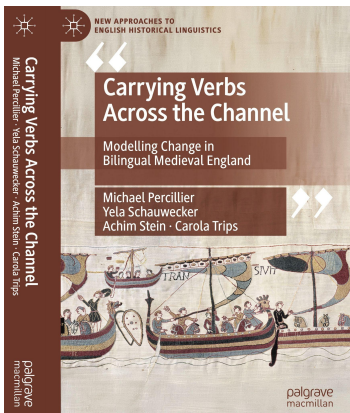
1. Historical linguistics: contact-induced change in medieval times

DFG project

Borrowing of Argument
Structure in Contact
Situations

([BASICS](#), 2015-2021)

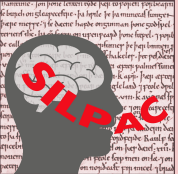
([Link to GEPRIS](#))



BASICS [Toolkit](#)

combines the software
tools and resources created
and used
in the framework of the
project.





1. Historical linguistics: contact-induced change in medieval times



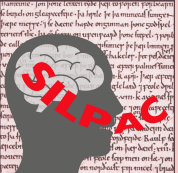
Welcome to BASICS!

A project funded by the DFG (2015-2021)

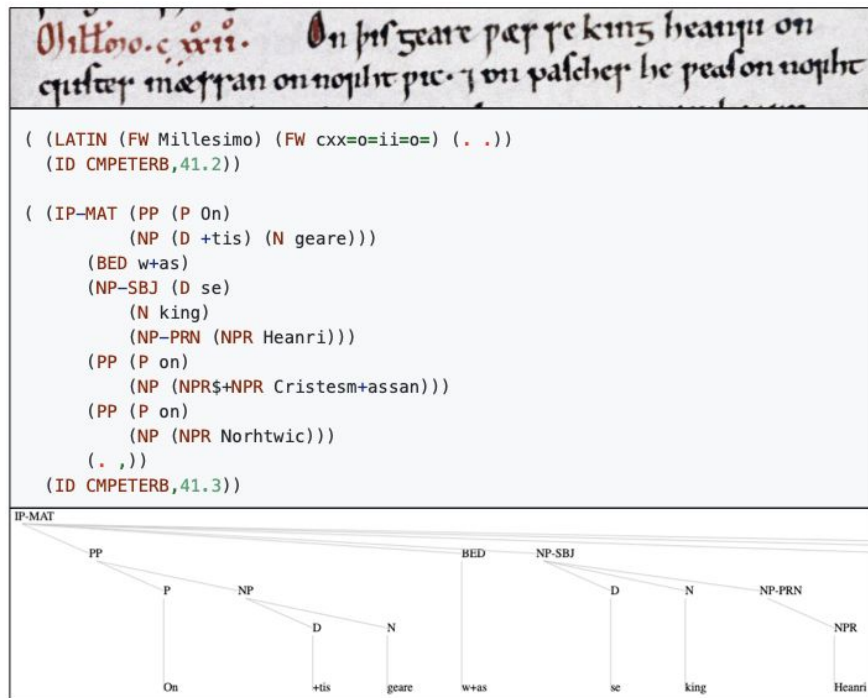
The project *Borrowing of Argument Structure in Contact Situations (BASICS)* investigates grammatical change in the medieval language contact situation between English and French which arose after the Norman Conquest (1066 until ca 1500). The German title is: *Entlehnung von Sprachkontaktsituation*

RQ: To what extent was the argument structure of Middle English (ME) verbs affected by the borrowing of Old French (OF) verbs, and with that English grammar?

Hypothesis: massive borrowing of OF verbs has at least accelerated grammatical change (if not initiated it).



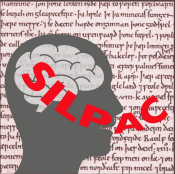
1. Historical linguistics: Corpus linguistics as method



← linguistic annotation:
part of speech tagging,
syntactic tagging,
syntactic parses

← tree structure

(illustration from Toolbox Anglistik <https://anglistik-toolbox.uni-mannheim.de/>)



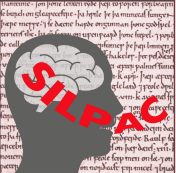
1. Historical linguistics: Corpus linguistics as method

We work with syntactically annotated Penn corpora of Middle English

- the [Penn-Helsinki Parsed Corpus of Middle English](#) (PPCME2, release 4); 1.2 mil. words, 56 texts
- the [Parsed Linguistic Atlas of Early Middle English](#) (PLAEME); c. 173,000 words, 68 texts
- the [Parsed Corpus of Middle English Poetry](#) (PCMEP), 216,649 words, 51 texts

and added additional layers of annotation:

- etymology of all lexical verbs (e= French/non-French origin)



1. Historical linguistics: Method

/~*

Sone come@l=comen@m=8522@e=nonfrench@ Merlyn unto the
kyng@l=king_Nlking@a=animate@
(CMMALORY,5.135)

*~/

/*

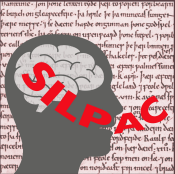
1 IP-MAT: 1 IP-MAT, 2 ADVP-TMP, 5 VBD, 7 NP-SBJ

*/

(0 (1 IP-MAT (2 ADVP-TMP (3 ADV Sone))
(5 VBD come@l=comen@m=8522@e=nonfrench@) ← lemmatized verb
(7 NP-SBJ (8 NPR Merlyn))
(10 PP (11 P unto)
(13 NP (14 D the) (16 N kyng@l=king_Nlking@a=animate@))))
(18 ID CMMALORY,5.135))

/~*

Lemmatisation of verb *comen*: @l=, MED-ID @m=, @e=nonfrench



1. Historical linguistics: Method

/~*

So departed@l=departen@m=11123@e=french@ Merlyon,
(CMMALORY,34.1096)

*~/

/*

1 IP-MAT: 1 IP-MAT, 2 ADVP, 5 VBD, 7 NP-SBJ

*/

(0 (1 IP-MAT (2 ADVP (3 ADV So))

(5 VBD departed@l=departen@m=11123@e=french@)

lemmatized verb



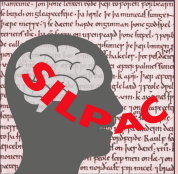
(7 NP-SBJ (8 NPR Merlyon))

(10 . ,))

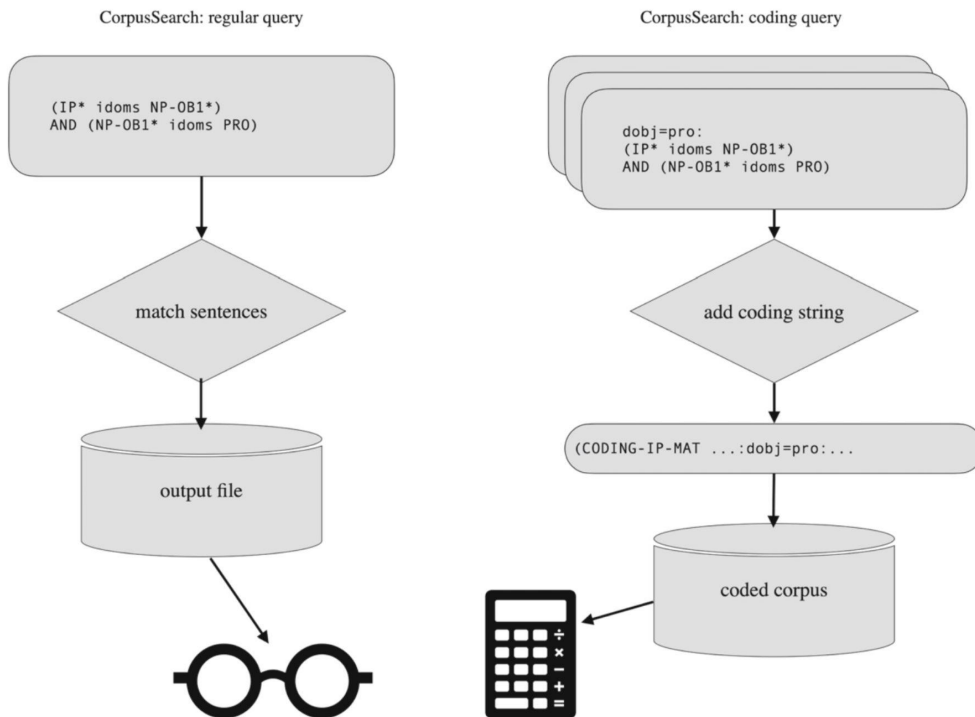
(12 ID CMMALORY,34.1096))

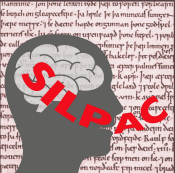
/~*

Lemmatisation of verb *departen*: @l=, MED-ID @m=, @e=french



1. Historical linguistics: method workflow





1. Historical linguistics: Method

outfile of
search



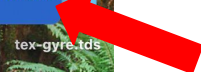
The screenshot shows a macOS desktop with a background image of redwood trees. In the foreground, there are three windows:

- Terminal Window:** Displays the output of a search command. It shows a list of words and their corresponding corpus search results. The search is for the word "malory" in the "combined-ME-corpora-v17" corpus. The results show that "malory" is found in the "combined-ME-corpora-v17" corpus, with a frequency of 16. The search is completed, and the output file is "ovinne-malory.out".
- File Browser Window:** Shows the contents of the "v2" directory. It lists several files, including "v2-30mf-verbs", "v2-m1-cmev17.out", "v2-m1-cmev17.q", "v2-m2-cmev17.out", "v2-m2-cmev17.q", "v2-m3-cmev17.out", "v2-m3-cmev17.q", "v2-m4-cmev17.out", "v2-m4-cmev17.q", and "v2-m4-cmev17.Kopie.q".
- Shell Window:** Displays the command "Ring-VL-unwrapping-the-mind --bash - 80x24" and the output of the search command. It shows the command "ovinne-malory.out" and the output "99% (37856,0) (corpus search) Wed Sep 24 11:59 2.12".

query files
and output
files



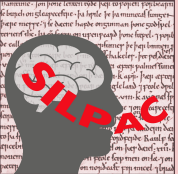
query
command
in shell





1. Historical linguistics: a case study of lability





1. Historical linguistics: a case study of labiality

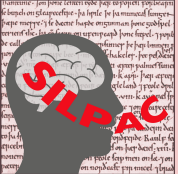
What is labiality/a labile verb?

A verb may describe the same situation in both **causal sense**, with a causal agent expressed by the subject and the patient/experiencer expressed by the object of the verb, and **inchoative sense**, where the verb excludes a causal agent and describes the situation as occurring spontaneously. (Haspelmath 1993, 90-91)

- (1) a. **Tom broke the window.**
- b. **The window broke.**

=> non-directed/labile: same form of verb is used for both argument structures

=> change of state verbs, the new state (**the broken window**) is a result



1. Historical linguistics: a case study of lability

Haspelmath (1993): English is quite unique in showing a strong preference for lability (also compared to the other Germanic languages)

- (3) a. She **sank** the boat.
b. The boat **sank**.
c. Sie **versenkte** das Boot.
d. Das Boot **(ver)sank**.

- (4) a. She **opened** the door.
b. The door **opened**.
c. Sie **öffnete** die Tür.
d. Die Tür **öffnete sich**.

Table 3. Expression types by language

	total	A	C	E	L	S	A/C	% non-dir.
Russian	31	23	0	5	0	3	46.00	26
German	31	14.5	0	4	11.5	1	29.00	53
Greek	31	13.5	0	0	16.5	1	27.00	56
Rumanian	30	24	1	0	3	2	24.00	17
French	31	20.50	2	0	7.5	1	10.25	27
Lithuanian	31	17.5	6	6	0.5	1	2.92	24
Hebrew	31	20.5	7.5	2	1	0	2.73	10
Arabic	31	17	8.5	3	1	1.5	2.00	18
Georgian	31	9	4.5	15.5	0	2	2.00	56
Armenian	31	16	8.5	5.5	0	1	1.88	21
Swahili	31	11	11	8	0	1	1.00	29
Finnish	28	12	13.5	0.5	0.5	1.5	0.88	9
Udmurt	31	10.5	12.5	4.5	2.5	1	0.84	26
Hungarian	31	7	9	12	0	3	0.78	48
Lezgian	31	8	12	6	5	0	0.66	35
Hindi-Urdu	31	7.5	14	7.5	2	0	0.54	31
Turkish	30	9	17.5	2.5	0	1	0.51	12
Mongolian	31	6	22	2	0	1	0.27	10
Indonesian	31	0	14	17	0	0	0.04	55
English	31	2	0	1	25	3		94
Japanese	31	3.5	5.5	20.5	0.5	1		71
total	636	243	164.5	128.5	69	310		

Abbreviations:

A = anticausative alternation

C = causative alternation

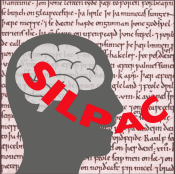
E = equipollent alternation

L = labile alternation

S = suppletive alternation

A/C = ratio of anticausative to causative pairs

% non-dir. = percentage of non-directed pairs



1. Historical linguistics: a case study of lability

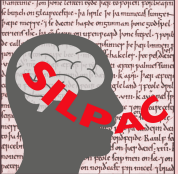
Old English (OE) showed few labile verbs and used morphology and stem form alternation (as other Germanic languages do) (cf. García García 2020).

stem vowel alternation/morphologically derived

- (5) a. *sencan* “sink” (transitive)
- b. *sincan* “sink” (intransitive)
- (6) a. *drygan* “dry” (transitive)
- b. *adrygan/adruwian* “dry” (intransitive)

labile change-of-state verbs (no change of verb form)

- (7) a. *baernan* “burn” (transitive/intransitive)
- b. *bigan* “bend” (transitive/intransitive)
- c. *godian* “improve” (transitive/intransitive)

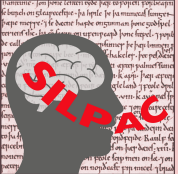


1. Historical linguistics: a case study of lability

In Middle English stem vowel alternations and morphologically derived patterns are lost leading to labile verb forms (8 a. and b.), some OE verbs remained labile (9 a.), some became labile (9 b.) and some are **copies from OF** (c.) (cf. Ingham 2020).

- (8) a. *sink* “sink”
b. *drien* “dry”

- (9) a. *breken* “break”
b. *cleven* “split”
c. *avauncen* “advance”



1. Historical linguistics: a case study of lability

How does language contact come in here?

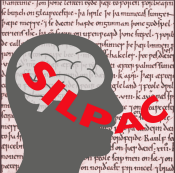
- The contact situation between Old French (OF) and Middle English (ME) in medieval times (ca 1150-1500) led to copying of verbs and as a result changes in the grammar of English (cf. Ingham 2012, Percillier et al. 2024)
- Ingham (2020) on lability:
 - Lability was the replication of argument structure properties of French verbs; when OF verbs were copied to ME **their argument structure was copied along (full set)**.
 - **OF/AN exhibited many labile verbs** (change of state/location, cf. Heidinger 2010)
 - between the end of the 12th up to the 14th century individual **bilingualism** existed which favoured grammatical replication



OF :	{ NP	_____	NP }
<i>rostir</i> 'roast'	Agent		Patient/Theme
<i>rouler</i> 'roll'	{ NP	_____ }	
etc.	Patient/Theme		

ME	:	{	NP	_____	NP	}
<i>rosten</i>			Agent		Patient/Theme	
<i>rollen</i>		{	NP	_____	}	
etc.			Patient/Theme			

28



1. Historical linguistics: a case study of labiality

ME rosten 'roast' copied from OF rostir 'roast'

OF: trans: [_{NP} il] **rosterent** [_{NP} phase] (=a sacrificial lamb) sur le feu (*Bible Royal* 300vb)
AGENT PATIENT

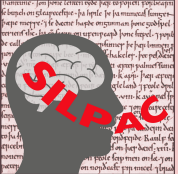
intrans: mettez [_{NP} le gars] sur un espé [...] e lessez **roster** molt bien ... (*Five Med MSS* 117.E1)
PATIENT

ME: trans: By a mykel fir [_{NP} he] sat, **Rostyng** [_{NP} a swyn gret & fat].
AGENT PATIENT

(a1450(a1338) Mannyng Chron.Pt.1 (Lamb 131)12342)

intrans: Turbot..when [_{NP} it] **rostis**, springle on salt.
PATIENT

(?a1475 Noble Bk.Cook.(Hlk 674)97)



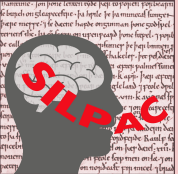
1. Historical linguistics: a case study of lability

Strong version of the contact hypothesis: French labile verbs "drive the expansion of lability" in English. They are always copied with both structures.

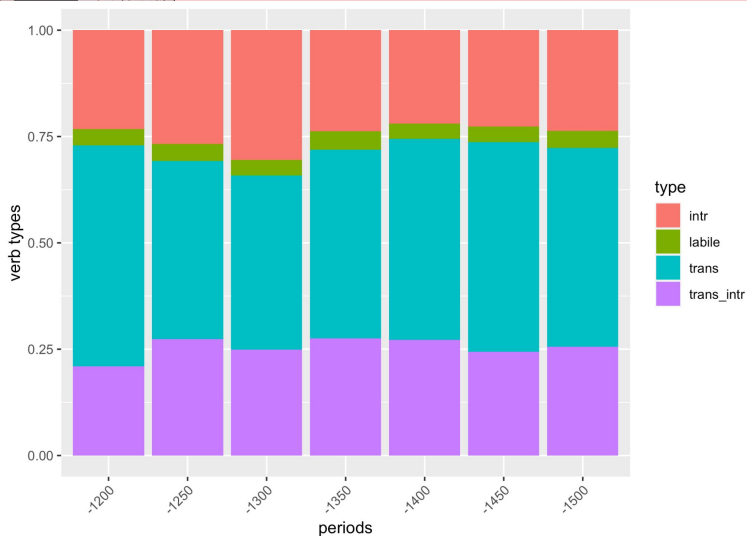
Alternative hypothesis: Lability, and syntactic alternations in general, are rule-governed, i.e. licensed and constrained by the event structure ([[x CAUSE [y BECOME STATE]]]; the structure is already there).

Based on our data we find that:

- French lability was not a sufficiently frequent cue for native speakers/learners → labile verbs don't "drive" change
- The English event structure was stable over time. Although French verbs formed a large proportion of the vocabulary, compared to English verbs they were not more liable to alternate.



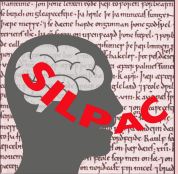
1. Historical linguistics: a case study of lability



period	trans	intr	trans/intr	labile	labile_ratio
-1200	387	173	156	29	3,89 %
-1250	785	502	513	74	3,95 %
-1300	530	395	323	48	3,70 %
-1350	654	351	406	64	4,34 %
-1400	662	308	382	51	3,64 %
-1450	634	291	313	47	3,66 %
-1500	532	270	292	46	4,04 %

- number of labile verbs is stable
- productive rule of lability already existed in OE
- OF contributed to the increase but did not initiate the change

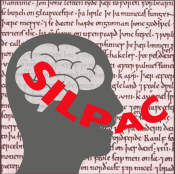
1. Automatic extraction of verbs occurring transitive or intransitive
2. min. 5 occurrences
3. **labile = manual verification**: causative-anticausative alternation (restricted to CoS verbs) attested in the combined ME corpora (PPCME2, PCMEP, PLAEME) or Middle English Dictionary (MED).
4. intr = always intransitive
5. trans = always transitive
6. trans_intr = both, but not labile



1. Historical linguistics: a case study of lability

Labile verbs copied from OF often **only show the causative transitive structure** in contrast to the native verb which already used to be labile in OE (event-structure governed lability)

- ME *curen* ('to cure' < OF *curer*)
 - **causative** 'to cure, restore to health': *He cured Naaman, þe prince of Surre', fro seknesse of lepre.* (PPCME2, CMCAPCHR,34.69)
 - **anticausative** not found in corpus
- ME *helen* ('to cure, heal' < OE *hælan*, labile) => compare German *heilen*
 - **causative** 'to cure, heal': *he tol=de is fader fore Hov a Naddre him hurte sore And hov Iesus hel=de him* 'he told his father how an adder severely hurt him and how Jesus healed him ...' (PLAEME, LAUD108AINFANCY.1154)
 - **anticausative**: *De seste is þe wunde þt eauer wursed on hond & strengere is to healen* 'the sixth is the wound that ever worsens in the hand and is stronger to heal' (PPCME2, CMANCRIW-1,II.242.3517)



1. Historical linguistics: bringing in acquisition

The alternative hypothesis is supported by language acquisition research (e.g. Scott & Fisher 2009).

- Child learners **do not receive sufficient evidence about verb lability** (alternations found in newspaper text are absent in child-directed speech).
- Despite this lack of evidence, children successfully acquire different verb alternations because:
 - they are sensitive to the semantics of verbs and argument roles (e.g. Causer, Patient)
 - they can detect if roles of subject and object overlap or not
 - they use these as **distributional features**

Scott & Fisher (2009):

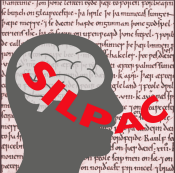
"Can young children use **distributional features** to assign meanings to novel verbs that occur in the causal and unspecified-object alternations?"

causal alternation
(= causative-anticausative
= P-lability):

x moves y → y moves

unspecified-object alternation:

x pushes y → x pushes



1. Historical linguistics: bringing in acquisition



Causal dialogue

A: Matt dacked the pillow.
B: Really? He dacked the pillow?
A: Yeah. The pillow dacked.
B: Right. It dacked.

Unspecified-object dialogue

A: Matt dacked the pillow.
B: Really? He dacked the pillow?
A: Yeah. He dacked.
B: Right. He dacked.

Contact-activity test event



Causal test event



Same-verb: "The girl is dacking the boy. Find dacking."

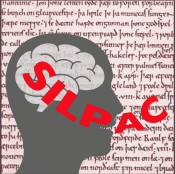
Different-verb: "The girl is pimming the boy. Find pimming."

Figure 2. Training and test phases for the novel verb (Experiment 2). Test events: Contact-activity event (left) and caused-motion event (right).

(Scott & Fisher 2009: 790)

The results "add to a growing body of evidence that young children represent verbs somewhat independently of the sentential context in which they occur" (Scott & Fischer 2009:795).

→ Acquisition of lability does not require sentence-structure cues (i.e., the presence of both structures in the input data). It is not acquired "in one package".



1. Historical linguistics: summary

Interim summary of case study

- Labile verbs have always been part of the English language, i.e. the event structure for these change of state verbs (causative-anticausative, resultative) has been part of the grammar of English before contact with OF
- Labile verbs copied from OF increased the number to some degree but did not drive the change. (answer to question 3)
- Labile verbs increased because phonological/morphological marking was lost
- **Cognitive aspect:** resultativity is acquired by children when they learn this type of verb, it becomes part of their event structure ([[x CAUSE [y BECOME STATE]])
- children use distributional features (e.g. subject animacy) to learn the alternation
=> the link between cognitive capability and language acquisition is constant across time (answer to question 1)

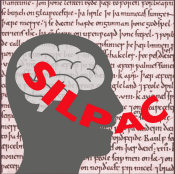


1. Historical linguistics

Interim summary of historical linguistics (my flavour)

Historical linguistics

- explains language change and identifies mechanisms of change
 - uses historical corpora to describe and analyse structural change
 - is theoretically informed and seeks to explain the problem of language change by seeing children acquiring language as drivers of change
- => this is the link to cognitive science!

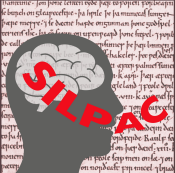


Your turn! How do you use *escalate*/*eskalieren*?

What about verbs that were copied from English into German like e.g. *eskalieren* (escalate)?

What does the verb *escalate* mean and how is it used in English (structures, i.e. trans./intrans.)?

What does the verb *eskalieren* mean and how do you use it in German (structures, i.e. trans./intrans.)?



escalate/eskalieren

Oxford English Dictionary Dictionary 🔍 Advanced search AI Search Assistant Signed in as Mannheim Univer...

ed 1972; not fully revised (entry history) Nearby entries

Share Cite Contribute Single page Tabbed

escalate

VERB

Factsheet Etymology Meaning & use Pronunciation Frequ...

2. *figurative (transitive and intransitive)*. To increase or develop by successive stages; spec. to develop from 'conventional' warfare into nuclear warfare.

1959-

1959 The possibility of local wars 'escalating' into all-out atomic wars'.

Manchester Guardian 12 November 1/1 ...

1961 Using tactical nuclear weapons which would be likely to escalate hostilities into a global nuclear war.

Economist 28 October 317/2 ...

1962 To keep the Communists from escalating the war to a stage in which they are capable of..taking over the reins of local government themselves.

Listener 29 March 543/2 ...

1962 Something short of a deliberate Soviet attack, some sort of accident, some sort of border incident, will escalate so rapidly into nuclear war that we shall be unable to stop it.

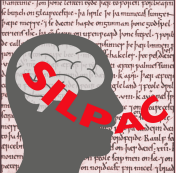
Listener 29 March 549/1 ...

1963 The wish of the author to magnify or escalate (favorite new word in Washington) the importance of a trivial utterance by grandiloquent terminology.

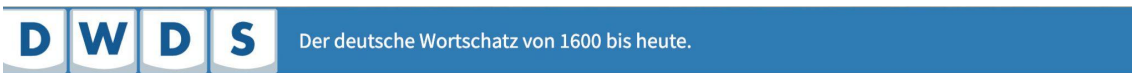
Horizon (U.S.) July 37/1 ...

1963 Peace as well as war, can escalate.

Oxford Mail 26 July 6/9 ...



escalate/eskalieren



[Startseite](#) / [Wörterbuch](#) / [eskalieren](#) – Schreibung, Definition, Bedeutung, Etymologie, Synonyme, Beispiele

eskalieren



eskalieren

[Lesezeichen](#)

[zitieren/teilen](#)

[+ ausklappen](#)

Grammatik Verb · **eskaliert**, **eskalierte**, **ist/hat eskaliert**

Aussprache

Worttrennung es-ka-lie-ren

Wortbildung mit ›eskalieren‹ als Erstglied: [↗ Eskalation](#) · mit ›eskalieren‹ als Letztglied: [↗ deeskalieren](#)

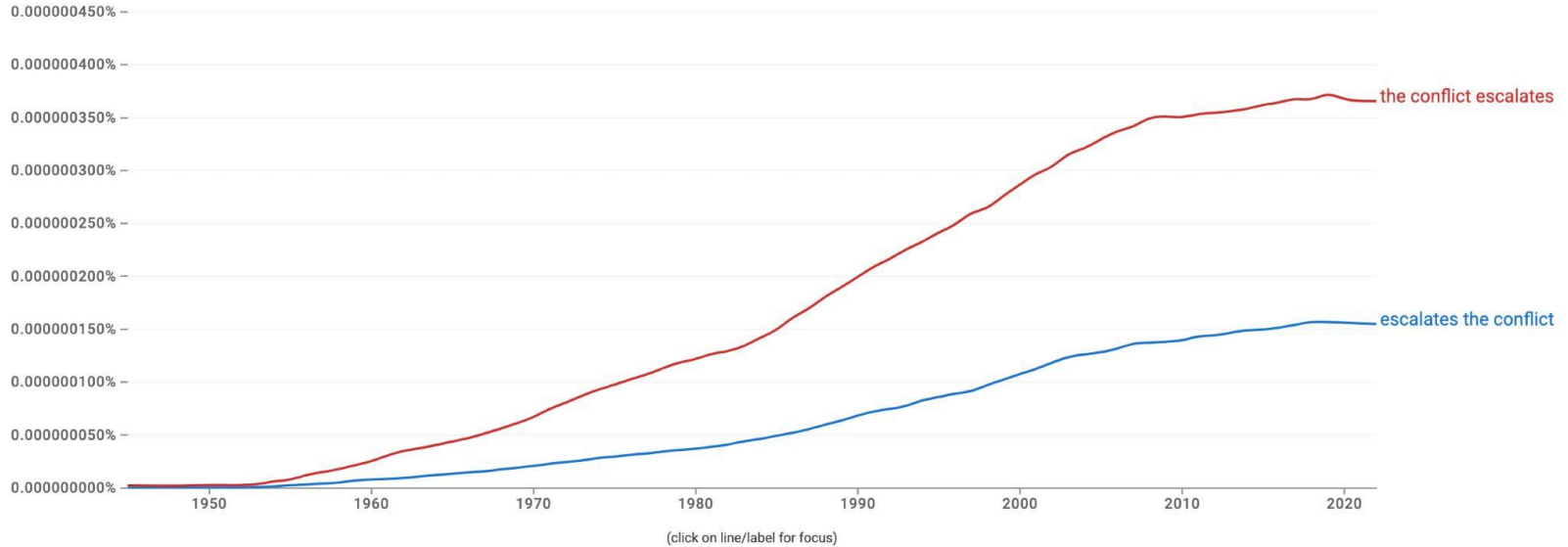
Bedeutungsübersicht

1. [bildungssprachlich] (etw. eskaliert) [sich im Ausmaß, in der Stärke steigern, sich ausweiten](#)
2. [bildungssprachlich] (jmd., etw. eskaliert etw.) [etw. im Ausmaß, in der Stärke steigern, ausweiten](#)

Und wenn die Situation **eskaliert** (= sich verschärft, zuspitzt), wird sich wie bisher schon die Europäische Zentralbank der Sache annehmen. [Die Zeit, 06.01.2014]
Die Angriffe aus der Luft **eskalierten** die Lage am Boden. Die Flüchtlingsströme drohten die ganze Region zu destabilisieren. [Die Zeit, 24.06.2009]

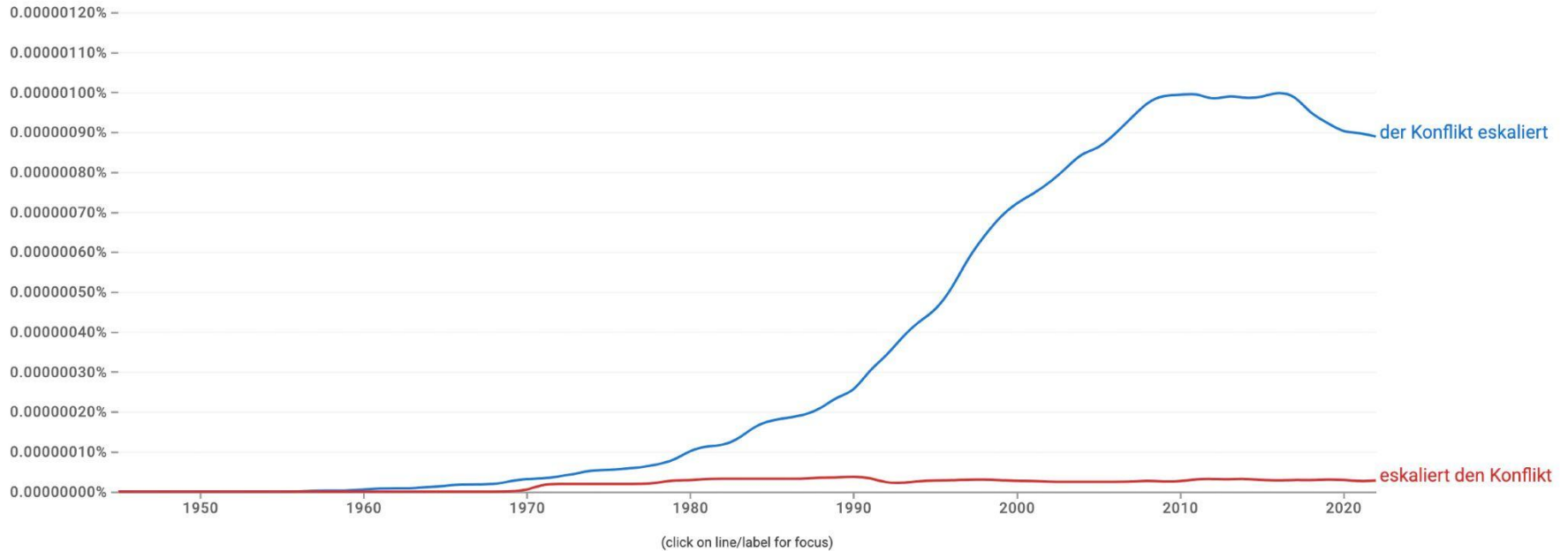


escalate/eskalieren





escalate/eskalieren





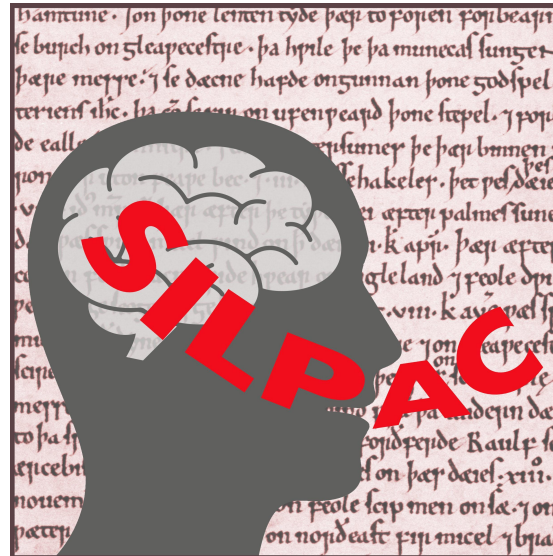
escalate/eskalieren

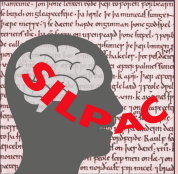
Maskenverweigerer eskalieren lautete kürzlich eine Schlagzeile in einer Tageszeitung. Der Leser stutzt, und dies mit gewisser Berechtigung. Das Verb *eskalieren* wurde in der zweiten Hälfte des 20. Jahrhunderts aus dem englischen *to escalate* entlehnt. Es gehört zum Wort *escalator* („Rolltreppe“), einer Bildung aus *to escalade* („eine Mauer, eine Festung mit Leitern erstürmen“). Zugrunde liegt das französische *escalade* („Erstürmung einer Mauer, einer Festung mithilfe von Leitern“), das wiederum zu lateinisch *scalae* („Leiter, Treppe“) gehört.

Eskalieren kann transitiv (also mit einem Akkusativobjekt) eingesetzt werden im Sinne von „steigern, verschärfen“: *Die Widerstandskämpfer eskalierten ihren Kampf bis zum Terror.* Intransitiv wird das Verb verwendet als Synonym für „sich verschärfen, sich ausweiten“: *Es besteht die Gefahr, dass die Auseinandersetzungen zwischen den Tarifpartnern eskalieren könnten. Die Demonstrationen gegen die verhängten Maßnahmen sind am Wochenende weiter eskaliert.* Im intransitiven Gebrauch des Verbs funktioniert die Perfektbildung übrigens sowohl mit *haben* als auch mit *sein*, wobei die Kombination mit *sein* die etwas häufigere ist.

Zurück zu unserer Schlagzeile: In der Umgangs- und vor allem auch Jugendsprache hat sich die Bedeutung des Verbs in letzter Zeit in eine andere Richtung entwickelt, es findet Verwendung im Sinne von „ausrasten, die Kontrolle verlieren“: *Mein Vater ist voll eskaliert. Maskenverweigerer eskalieren.*

2. Psycho-historical linguistics

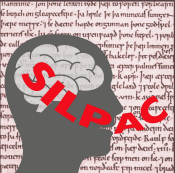




2. Psycho-historical linguistics

Research unit "Structuring the input in language processing, acquisition and change" (SILPAC, FOR 5157 funded by the DFG)

- aims to understand language change from the cognitive perspective and focusses on the **process underlying language change**
- **claim:** changes in the structure of the input lead to changes in processing which in turn leads to changes in the output which serve as input to new learners (new grammar). These grammars are the result of acquisition that we see in historical texts.
- This is why it is key to investigate processing
=> psycholinguistically-driven approach to language change



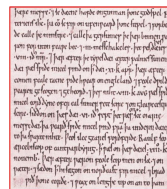
2. Psycho-historical linguistics



H → P approach

$$\theta_N = \frac{N}{\ln N}$$

The Tolerance and
Sufficiency Principles



Historical corpora

New E-language:

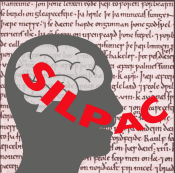
- changes in the structure of the input
- initial locus of change

Changes in processing:
→ implicit learning
→ adaptation

New I-language:
acquisition
learnability

**Grammar of
historical texts**
=
result of
acquisition

**Psycholinguistics as missing link
in theories of language change**



2. Psycho-historical linguistics

Modelling change in diachrony using the TSP

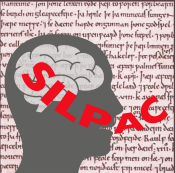
- frequency-based learning
- small early child lexicon
- precise quantitative-based predictions on the basis of corpus-data (is a rule productive or not?)

$$\theta_N = \frac{N}{\ln N}$$




Applied to the distributional learning of modal verbs in
Present-Day English and in Middle English
"Distributional Learning and Grammaticalization:
Modals in the History of English" (Yang & Trips 2025)





2. Psycho-historical linguistics

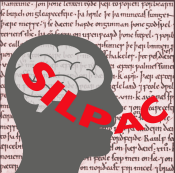
How do (small) children learn syntactic categories via distributional learning?

They use a learning principle called the **Tolerance Principle** (TP, Yang 2016)

A rule defined over N items can generalize if e , the number of the items that do not follow rules, does not exceed a threshold, $\theta N = N/\ln N$.



The TP as a mechanism of category formation: If a set of words is *sufficiently* homogeneous with respect to their distributional properties, then the set can be regarded as members of a single syntactic category. The TP threshold provides a precise value of what counts as “sufficiently” homogeneous.



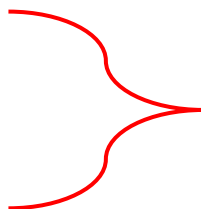
2. Psycho-historical linguistics

How do children identify modal verbs among the set of verbs?

I *do, am/m', have, will/'ll, know, go, can, think, want, see* + not/n't → I-followers +
not-predecessors

=> **10 most frequent words + predecessors (not)** in CHILDES database (MacWhinney 2000)

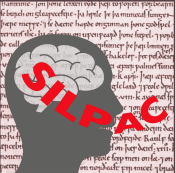
- (10) a. She **must** **not** **leave** right now.
b. She **has** **not** **left** yesterday.
c. She **leaves** now.
d. She did **not** **leave**.



V/V not
not V

auxiliary verbs: occur **left** of **not**

lexical verbs: occurs **right** of **not**



2. Psycho-historical linguistics

Out of 10 items, only 5 are robustly attested as "NEG predecessors":

do, will/'ll, am/'m, can, have



[+ NEG predecessor]

think, know, want, see, go



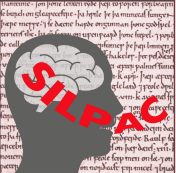
[- NEG predecessor]

Children can't build one homogeneous class of verbs, they subdivide into the sets of functional and lexical verbs; the TP threshold predicts this ($\theta_{10} = 4$)

What's the next element? A verb in the bare infinitive (+/- inflection)

modals: take the bare infinitive (She **must not leave** right now)

auxiliaries: don't take the bare infinitive (She **has not left** yesterday)



2. Psycho-historical linguistics

Two distributional profiles

- NEG preceding
- bare infinitive taking

are *sufficient* for a child to learn the three sets of verbs in English (they are enough to learn them).

Both are needed to learn them based on **positive evidence**.

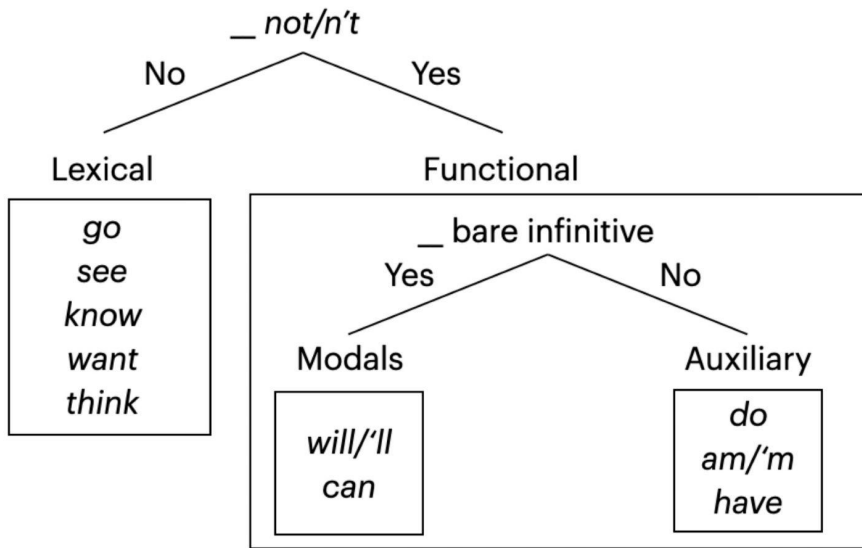


Figure 1

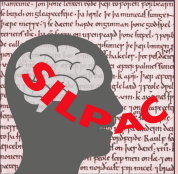
A decision tree that represents the major verbal categories that can be constructed on linear structures for a very small lexicon.



2. Psycho-historical linguistics

1. How did the historical child (Middle English) confronted with input that was undergoing change identify modals as a syntactic category?
2. Did they use the same distributional profiles?





2. Psycho-historical linguistics

Changes in negation system and placement of finite verbs:

ne and contracted ***ne*** as sentence negation

- (11) wið-uten hem **ne** cumst ðu ðar naure!
without them NEG come you there never
‘Without them you will never get there!’

(M1, a1225, VICES1,25.265)

In al denemark **nis** wimman So fayr so sche, bi seint iohan!
in all Denmark NEG-is woman so fair so she by Saint John
‘In all Denmark there is no woman as fair as she is, by Saint John!’

(M2, 1290, Havelok,47.1715.807)

(Yang & Trips 2025: 17)





2. Psycho-historical linguistics

Changes in negation system and placement of finite verbs:

ne as sentence negation, negative concord, and verb movement to left of ***not***

(12) Certes, he that is fool-large **ne** yeveth **nat** his catel, but
certainly he that is foolishly-generous NEG gives NEG his property but
he leseth his catel.

he loses his property

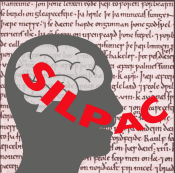
‘Certainly, he who is improvident doesn’t give his property unless he loses it.’
(M3, c1390, CTPARS,316.C1.1189)

(13) Prowde men or women lufes **noght** stalworthly;
proud men or women love NEG vigorously
‘Proud men and women don’t love vigorously.’

(M4, 1450, ROLLEP,112.847)

(Yang & Trips 2025: 17)





2. Psycho-historical linguistics

The comparability of child and historical data

- 10 most frequent verbs in all corpora: auxiliaries *have, be, do*, modals *will, can, shall*
- 20 most frequent verbs in all corpora: verbs of motion (*come, go*), verbs of saying (*say, tell, speak*), verbs of cognition (*see, look, know, think*) and verbs of transfer of possession (*give, take*).
- a substantial semantic overlap of the most frequent verb types a child is exposed to in all three corpora (Kodner 2019)

(Yang & Trips 2025: 14)

BNC, spoken part			CHILDES			ME combined corpora		
rank	lemma	freq	rank	lemma	freq	rank	lemma	freq
1	be	589675	1	be	11305	1	ben 'be'	57089
2	have	203372	2	do	10243	2	haven 'have'	15071
3	do	171766	3	go	4808	3	shulen 'shall'	13416
4	get	95491	4	do neg not	3969	4	seien 'say'	9487
5	go	71338	5	have	3534	5	mouen 'may'	7967
6	say	63330	6	get	2973	6	don 'do'	7868
7	know	61719	7	see	2301	7	comen 'come'	6379
8	will	56883	8	know	2103	8	willen 'will'	5921
9	think	52349	9	put	1951	9	maken 'make'	4847
10	can	49920	10	think	1901	10	sen 'see'	3810
11	would	46242	11	can	1732	11	taken 'take'	3112
12	see	32843	12	come	1546	12	yeven 'give'	2758
13	come	31685	13	say	1394	13	wenden 'go'	2350
14	mean	26161	14	will	1314	14	witen 'know'	2026
15	want	23950	15	want	1286	15	gon 'go'	1996
16	take	20836	16	look	1245	16	tellen 'tell'	1809
17	look	20496	17	make	1115	17	thinken 'think'	1770
18	could	20148	18	take	1058	18	speken 'speak'	1742
19	make	19656	19	like	1006	19	bringen 'bring'	1599
20	put	18908	20	tell	952	20	leten 'let'	1580

Table 1
Comparability of data from the BNC, CHILDES and Penn ME corpora in terms of verb frequency



2. Psycho-historical linguistics

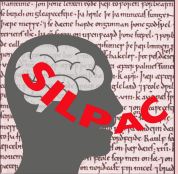
How do historical children identify modal verbs among the set of verbs?

sbj pron – ***ben/have/mouen/shulen/willen/comen/don/maken/seien/sen*** → *sbj pron-followers*

=> **10 most frequent words** in Middle English corpus data

Differences to Present-Day English:

- *ne* is the sentence negation that is also contracted with auxiliaries and modals but never with lexical verbs
- *ne* is lost by 1420 and *not* increases and becomes sentence negation
- lexical verbs still occur in a higher position (left of negation)



2. Psycho-historical linguistics

Comparison of distributional profiles in ME and PDE

	Distributional profiles	
	“Neg”	“bare infinitive”
early Middle English	+ (<i>ne</i>)	+
late Middle English	-	-
early Early Modern English	-	-
late Early Modern English	+ (<i>not</i>)	+
Present-Day English	+ (<i>not</i>)	+

Table 6
Distributional profiles in ME, EModE and PDE for modals

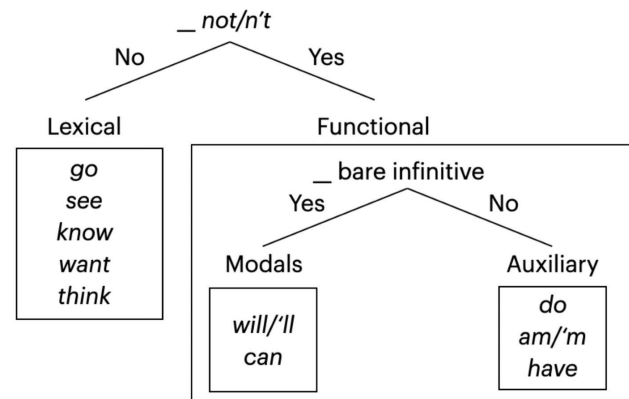


Figure 1
A decision tree that represents the major verbal categories that can be constructed on linear structures for a very small lexicon.

(Yang & Trips 2025: 9, 26)



2. Psycho-historical linguistics

Summary

- for PDE the two distributional profiles **_not/n't** and **_bare infinitive** are sufficient for children to learn the category of modal verbs
- for the beginning of the ME period **contracted *ne*** and **_bare infinitive** are profiles
- between late ME and late Early Modern English (end of 18th) only the profile **_bare infinitive** exists but it isn't enough to learn the category of modals.
- only when verb movement is lost does **_not/n't** become a profile that together with **_bare infinitive** children can use to learn this class of verbs.

=> on the basis of frequency trimming and distributional learning (discovery procedure) that is determined by the TP we can model the historical child.



3. Future perspectives

- To model the results gained from this study by means of an **artificial language learning experiment**
- Miniature language that learners have to learn on the basis of a lexicon of pseudowords

*El*_[determiner] *bone*_[noun] *mip*_[auxiliary] *ke*_[negation] *tan*_[verb] *el*_[determiner] *gass*_[noun]

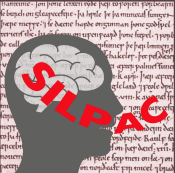
- Condition 1: both distributional profiles are reliable to learn the class of modals
- Condition 2: only one of these profiles is reliable to learn the class of modals





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Appendix



Linguistics: how do learners acquire language?

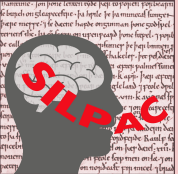
Language acquisition: process-oriented concept

- dynamic, step-by-step way
- action of learning
- stages of development
- environmental factors

Language learning: theory-oriented concept

- Is a language learnable by a human mind?
- What are the necessary conditions for a language to be learned?
- What are the innate cognitive abilities that make this possible?





1. Historical linguistics: language acquisition and language change

"A central problem of linguistic theory has been the **logical problem of language acquisition**. The standard framework of linguistic theory models language acquisition as a search for **the correct target grammar** from a class of available target grammars on the basis of example sentences. Psychological learning theory imposes several constraints on the kinds of algorithms employed for this search. Linguistic theory imposes constraints on the class of available grammars that need to be searched to solve the problem. Over the last few decades, significant attempts have been made to computationally model the phenomenon of language acquisition — and every effort has been made to explain acquisition — to ensure **learnability** — in short, to make sure that the logical problem of language acquisition can be solved."

(Niyogi and Berwick, 1998, 192)



1. Patient-Lability/the causative-anticausative alternation

- Lability is associated with the event structure (ES) of alternating verbs (Pinker 1989, Levin 1993)
- the ES of alternating verbs consists of a causing and a resulting event.
- the anticausative construction drops the causing event and is inhibited if semantically relevant components like Manner or Control are linked to the Causer.

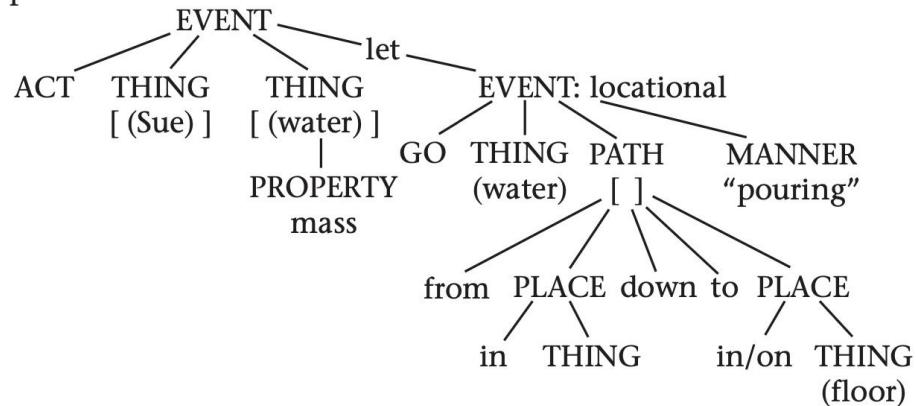
- (2) a. *move*: [x CAUSE [y MOVE]]
b. *pour*: [x CAUSE [y MOVE [by POURING]]]
c. *push*: [x CAUSE [y MOVE] [by PUSHING]]

⇒ The structure of the CAUSE event constrains the alternation.

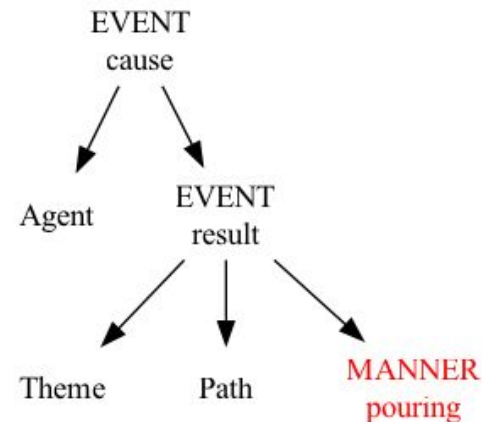


1. Patient-Lability/the causative-anticausative alternation

pour:



simplified



Verbs of force exertion such as *push* or *shove*, [...] always specify a **manner** within the ACT structure

(Pinker 2013 [1989], 271)

